GitaVerse: The Ultimate Solution

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Abstract: In this paper, we have proposed and implemented the functionality of the Gita Bot. This groundbreaking solution draws inspiration from the insights of the Bhagavad Gita to address the prevalent issue of stress in today's fast-paced world. The Gita Bot serves as an AI assistant, offering tailored insights, tips, and mindfulness exercises to help users manage stress effectively. By leveraging artificial intelligence and natural language processing, users can engage in meaningful conversations that promote resilience and deeper understanding of stressors. We detail the conception, development, and utilization of the Gita Bot in this study, supported by user reviews and quantitative assessments of stress levels before and after engagement. Our findings underscore the potential of combining modern technology with traditional wisdom to support mental well-being in the digital age.

Keywords— Gita Bot, Bhagavad Gita, stress management, artificial intelligence, mindfulness, mental well-being

1. Introduction

It is evident that the majority of individuals are tense and anxious. Apart from that, let's merely discuss Indian professionals who are engulfed in a great deal of worry and anxiety. Since AI arrived and widespread layoffs took over, leading to job losses and suffering for our experts, particularly in the tech sector, we all know how crucial moneymaking is in India, where meeting basic necessities is a serious competition. When faced with such issues, the Bhagavad Gita can be of great assistance.

When confused warrior Arjuna asked for help from his charioteer, Shri Krishna about the dilemma of war against his own family, Shri Krishna stated some philosophical contexts which are still a solution to every individual's life problem.

All the stress at work in your life can take you down and stop your soul from doing good to yourself and others. To be precise this book talks out the ways to get rid of stress and make you a better person.

The scripture holds solutions to one's every life problem. It is considered a spiritual dictionary by the father of the nation "Mahatma Gandhi" and this book has been a source of inspiration for many leaders in India's independence movement.

Once a wise man said, "Difficulties are part of life, if one can bear the pain of having difficulties, tackling the pain right, life would be much better." Consider the ocean as an example. When you swim in it, the waves pull you down, and the bigger the wave, the more you are dragged down. However, what would happen if the swimmer dove under the waves? Eventually, the wave will pass and you won't be pulled beneath the water. The same is true with life; if you know how to handle a situation, it will eventually go away. Problems never remain forever.

1.1 The Purpose and Significance of Research

Today, the world of business has been more hectic and problematic than ever. The corporate sector has gone through tremendous forms of low productivity, there's a lack of morale, motivation is low in employees, even unfitting skills of leadership exists and there are so many other factors creating a problematic wave inside this world of corporate. (Adamopoulou, & Moussiades, 2020), (Rooein & Donya, 2019)

Additionally, these statistics demonstrate the reason for stress levels at workplaces are caused due to overtime, interference of personal relationships at work, lack of support from managers, etc. These types of things are inhumane and shouldn't be done to grow a company or a sector. Bhagavad Gita is believed to have the solution to all mankind's problems. The epic Mahabharata, a 2200-year-old book has a soulful possession in the business scenario, which is still relatable in the modern world.

1.2 Literature Review

The key teachings of the Bhagavad Gita related to stress management. According to Bhagavad Gita, there are 4 steps for attaining emotional stability, namely:

1.) Sense management: (gunas)
2.) Desire management: (kama)
3.) Anger management: (krodha)
4.) Mind management:(manas)

Step 1: Sense Management (gunas): The goal is to sublimate your entire low or negative energy system to a higher or positive energy system. Let's dive into the depth of the Bhagavad Gita shlokas (context/verses). Bhagavad Gita states, that a person who is willing to attain perfection on emotions, can still struggle even if there's discipline and knowledge. The impulses and desires can still drag a person down to the path of distractions. Take the example of a ship, the wind has the power to change its direction, similarly, our senses and desires can still sweep us away if we aren't careful (Sivakumar, 2022)(Prabhupada, 2006).

Then what's the way out? How to get rid of this trouble?
Bhagavad Gita states, that one should retreat their senses/desires completely from the dilemma they are suffering. Learning from a tortoise drawing its limbs to protect itself from predators.Therefore, in order to experience the peace, the calmness, first step should be controlling the senses(Sivakumar, 2022)(Prabhupada, 2006).

Let's get straight into how to master senses.

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Senses are the combination of these vital traits, namely, knowledge and calmness (sattva), desires (rajas), and laziness (tamas). They consistently account for one another and are present in different proportions in every individual. These three are the main elements that define an individual's personality. Just like the window has to be clean and clear to see the view vividly, similarly senses are human's window through which we see the world, which has to be clear.

Bhagavad Gita states that this purification of senses is an individual's journey from a prevailing tamarisk existence to a rajasic existence and from a primary rajasic state to a satavik state of existence. (Sivakumar, 2022)(Prabhupada, 2006)

The process of attaining such a level is stated as karma yoga. That's how we learn the purification of senses and desire with the help of the Bhagavad Gita. It's all about perspective. To achieve the state of emotional stability, master the first step by controlling the senses.

Step 2: Desire Management: (kama). According to our Indian culture, desires are classified into two categories:

a.) Desires directed by ethics (positive)

b.) Desires not directed by ethics (negative)

It states that satisfying a desire does not necessarily mean that it'll be extinguished but it'll intensify. It is the same as adding ghee to a Havana, adding ghee will flame the fire more. Similarly, the desires can never be satisfied enough. Bhagavad Gita states that one's desires are the only biggest hurdle to getting emotionally stabilized and realizing spiritually.

Gita states that the locus of our desires is present in sensing organs, the intellect and our mind. Desires are born with the nourishment of senses. As the senses are the root cause, the Bhagavad Gita guides us to master the control over senses, eventually it'll help in lowering the desires and will flourish the soul with satisfaction. Hence, this management is considered to be the 2nd step in the purification of self and helps in achieving supreme emotional stability.

Step 3: Anger Management (krodha). Anger is said to be a destructive force, anger can cloud thoughts, and the power to make clear decisions, hence resulting in downfall and deterioration of thoughts. Because this element (anger) can destroy the self, it has to be demolished.

Bhagavad Gita even states that underachieved desires are the main cause of anger. Therefore, anger must be controlled by working on your senses and likewise affecting desires for good. The gateway for attaining emotional stability can be achieved by removing these upper three elements from one's self.

Step 4: Mind Management (manas). The intellectual sensing organ is our mind. according to Bhagavad Geeta, an individual has two strong forces namely, attachment and antipathy(dvesha) continuously tackling our mind, resulting in struggling with peace. Anyone having a controlled and developed mindset can make friends, and enemies if not created. Now, the question is, how do we develop and control this state of mind?

Bhagavad Geeta answers, by consistently practicing (abhaysa) the art of detachment (vairagya) from senses. The learnings of Bhagavad Geeta align with the deep connection between the mind and senses. It warns that the mind of any positive individual who is seeking perfection can get corrupted easily by the effect of these senses, as they are a strong intellectual force.

According to previous learnings, it states that controlling senses leads to controlled desires and controlled desires help in anger management. It is obvious that senses are the core of mind management, hence the fourth step towards attaining emotional stability.

2. Architecture

Emphasizing a chatbot needs to understand the chatbot's purpose for both developers and end users. For developers, all the algorithms, tools, and structures have to be on point. To make an efficiently intended chatbot. For the end user, it is important what to expect from it so that the functionalities will be met.

2.1) Elements that are essential for designing and performing chatbot include

1.) Proper representation of knowledge.
2.) Strategy to generate solutions.(Chopra &Arora, 2024)(Chopra & Khurana, 2023)
3.) A collection of predefined answers that can be given when user input is not understandable to the chatbot.

The primary goal of making a system is to break it down into several modules to attain a modular development approach. The process initiates with the user's input, e.g. "What is the meaning of life?"

The model uses text messenger apps like Slack, Facebook, etc., or a voice assistant like Amazon Echo. After getting user input, the language-recognizing component translates the text or vocals to the best interpretation it can. The first thing to process is giving a natural understanding answer to the user. Either it can give the solution to the user's input query directly from the database chatbot's knowledge, or it can access the knowledge from internet websites through APIs.

Now, as the chatbot understands what the input is, it uses a protocol called, (the RGC) Responsive Generation Component to come up with a reply that will sound like how the user will interact (it's a human sounding text). It uses a particular technique called Natural Language Generation(NLG) which makes the response neutral to human user. (Adamopoulou & Moussiades, 2020).

2.2) Ways through which a chatbot can provide such responses

1.) **Rule-based.** This involves following predefined rules. For example: the user asks for the weather, and the chatbot might respond with a prewritten answer.
2.) **Retrieval bases.** The solution to a problem must be responded to from the information present in the database of the model.
3.) **Generative model.** This approach includes the generation of a solution from scratch based on the input of the user, often using ML algorithms.

In addition, there's another part called Dialogue Management Component that keeps track of entire user interaction, this helps in giving more accurate answers and it also learns for future conversations, and is likely to give more user-friendly and understanding solutions to problems (Tamrakar & Wani, 2021).

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3. Bhagavad Gita Chatbot: Proposed Work

3.1 Training

Bhagavad Gita contains dialogues and teachings between lord Krishna and prince Arjuna. The chatbot has to reply to the user's query/problem based on the conversations that are reserved in this spiritual book in the form of verses.
Pre-training the chatbot with a corpus of verses written in Bhagavad Gita. Retrain the model with the explanation of those spiritual texts to refine and purify the responses. This ensures a better solution to every question/query. Keeping track of conversations of user queries related to Bhagavad Gita. And using feedback and interactions to monitor and update the chatbot for future use.

3.2 Connect The Bhagavad Gita Chatbot To A Channel

Integrating: Deploying chatbots onto websites, spiritual forums, and social media platforms by machine learning algorithms, so that it only reaches those who are seeking spiritual content and sometimes engage in spiritual discussions.

3.3 Conversational Mode Of The Bhagavad Gita Chatbot

Spiritual Persona Enhancement: Building a chatbot model, which consists of the whole essence of Bhagavad Gita's learning, so that the user can experience a fostering answer to their life and spiritual problems.
Engaging User Experience: When people interact with chatbots, they have to experience a spiritual and meaningful experience. We need to make a chatbot that provides enlightenment to users, it has to engage users in conversations that will inspire them to understand and know about Bhagavad Gita's teaching better.

4. Building a Customable Model for Gitabot

4.1 A Problem-Solving Tool That Answers from the Bhagavad Gita

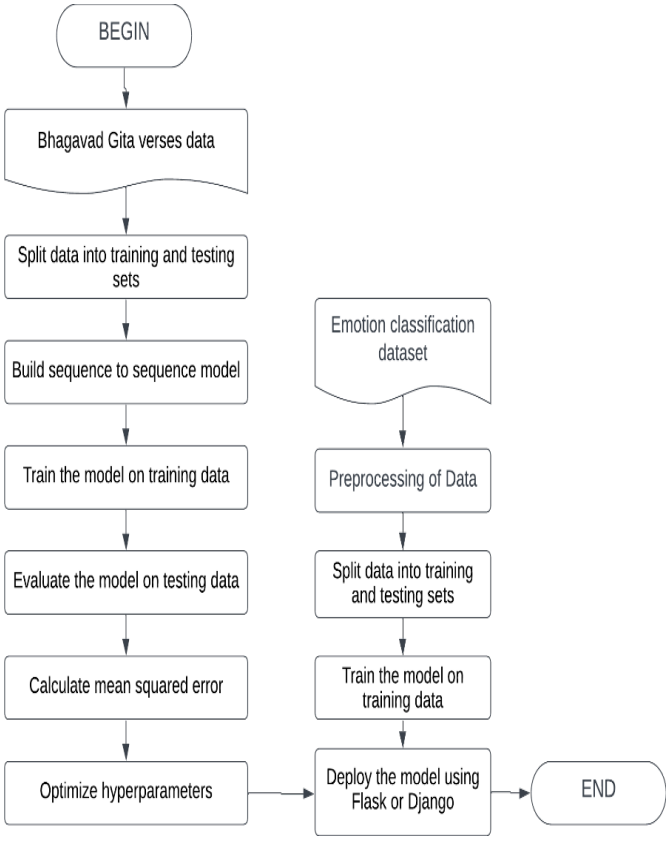


Fig. 1. Flowchat of Gitabot

4.2 Proposed Algorithm

INPUT: User Query, Bhagwad Gita Dataset, Emotion Classification Dataset

OUTPUT: Solution to users query personalized with the insights of Bhagwad Gita along with the analysis of user's emotional state

STEP 1: BEGIN

STEP 2: Load and preprocess Bhagavad Gita verses data

```
function preprocess_text(text): [8]
    tokens = word_tokenize(text)
    tokens = remove_stopwords(tokens)
    tokens = lemmatize(tokens)
    return ' '.join(tokens)
function remove_stopwords(tokens): [10]
    filtered_tokens = []
    for token in tokens:
        if token.lower() not in stopwords.words('english'):
            filtered_tokens.append(token)
    return filtered_tokens
function lemmatize(tokens):
    lemmatized_tokens = []
    for token in tokens:
        lemmatized_tokens.append(nlp(token)[0].lemma_)
    return lemmatized_tokens
```

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```
STEP 3: Split data into training and testing sets
        x: Features are the attributes and variables extracted from the
        dataset which are used as inputs to the model during training.
        y: Labels are the output or the target variable.

STEP 4: Build sequence-to-sequence model
        function build_sequence_to_sequence_model():
            model = Sequential()
            model.add(Embedding(input_dim=10000,
        output_dim=128, input_length=max_length))
            model.add(LSTM(64, dropout=0.2))
            model.add(Dense(64, activation='relu'))
            model.add(Dense(1, activation='sigmoid'))
            model.compile(loss='binary_crossentropy',
        optimizer='adam', metrics=['accuracy'])
            return model

STEP 5: Train the model on training data
        function train_model(model, X_train, y_train, X_test,
        y_test):
            model.fit(X_train, y_train, validation_data=(X_test,
        y_test), epochs=10, batch_size=32)

STEP 6: Evaluate the model on testing data
        function predict_answer(model, question):
            answer = model.predict(question)
            return answer

STEP 7: Calculate mean squared error
        function calculate_mean_squared_error(model, X_test,
        y_test):
            predictions = model.predict(X_test)
            mse = mean_squared_error(y_test, predictions)
            return mse

STEP 8: Optimize hyperparameters
        function optimize_hyperparameters():
            params = {'epochs': [5, 10, 20], 'batch_size': [32, 64, 128],
        'dropout': [0.2, 0.5]}
            grid_search = GridSearchCV(estimator=model,
        param_grid=params, n_jobs=1, cv=3)
            grid_search.fit(X_train, y_train)
            return grid_search.best_params_

STEP 9: Emotion Classification of User Query
        Import_pandas_NumPy_seaborn_libraries():
            Import the required libraries: pandas, NumPy, and
        seaborn
        Preprocess_data_using_NLTK():
            Tokenize the text
            Remove stop words
            Perform lemmatization or stemming
        Split_data_into_train_and_test_sets():
            Split the preprocessed data into training and testing sets
        Train_emotion_classification_model():
            Train the classification model using the training data

STEP 10: Deploy the model using Flask or Django
        function deploy_model(model):
            @app.route('/ask', methods=['POST'])
            def ask():
                question = request.get_json()['question']
                answer = predict_answer(model, question)
                return jsonify({'answer': answer})

STEP 11: END
```

4.3 Dataset Used

a. The dataset has 34792 records in total.

	Emotion	Text
0	neutral	Why ?
1	joy	Sage Act upgrade on my to do list for tommorow.
2	sadness	ON THE WAY TO MY HOMEGIRL BABY FUNERAL!!! MAN ...
3	joy	Such an eye ! The true hazel eye-and so brill...
4	joy	@lluvmiasantos ugh babe.. hugggzzz for u .! b...
	Emotion	**Text**
34787	surprise	@MichelGW have you gift! Hope you like it! It'...
34788	joy	The world didnt give it to me..so the world MO...
34789	anger	A man robbed me today .
34790	fear	Youu call it JEALOUSY, I call it of #Losing YO...
34791	sadness	I think about you baby, and I dream about you ...

Fig. 2. Dataset Used

b. Count of entries for every emotion.

```
joy         11045
sadness      6722
fear         5410
anger        4297
surprise     4062
neutral      2254
disgust       856
shame         146
Name: Emotion, dtype: int64
<AxesSubplot:xlabel='Emotion', ylabel='count'>
```

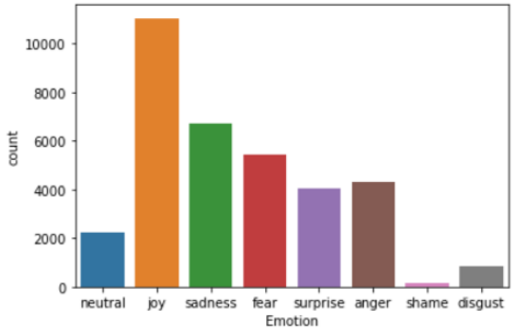


Fig. 3. Count of emotions

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c. Eliminating the stopwords from the dataset.

	Emotion	Text	Clean_Text
0	neutral	Why ?	?
1	joy	Sage Act upgrade on my to do list for tommorow.	Sage Act upgrade list tommorow.
2	sadness	ON THE WAY TO MY HOMEGIRL BABY FUNERAL!!! MAN ...	WAY HOMEGIRL BABY FUNERAL!!! MAN HATE FUNERALS...
3	joy	Such an eye ! The true hazel eye-and so brill...	eye ! true hazel eye-and brilliant ! Regular f...
4	joy	@Iluvmiasantos ugh babe.. hugggzzz for u .! b...	ugh babe.. hugggzzz u .! babe naamazed nga ako...
...	...	...	...
34787	surprise	@MichelGW have you gift! Hope you like it! It'...	gift! Hope like it! hand wear ! It'll warm! Lol
34788	joy	The world didnt give it to me..so the world MO...	world didnt me..so world DEFINITELY cnt away!!!
34789	anger	A man robbed me today .	man robbed today .
34790	fear	Youu call it JEALOUSY, I call it of #Losing YO...	Youu JEALOUSY, #Losing YOU...
34791	sadness	I think about you baby, and I dream about you ...	think baby, dream time

34792 rows × 3 columns

Fig. 4. Generating Clean Dataset

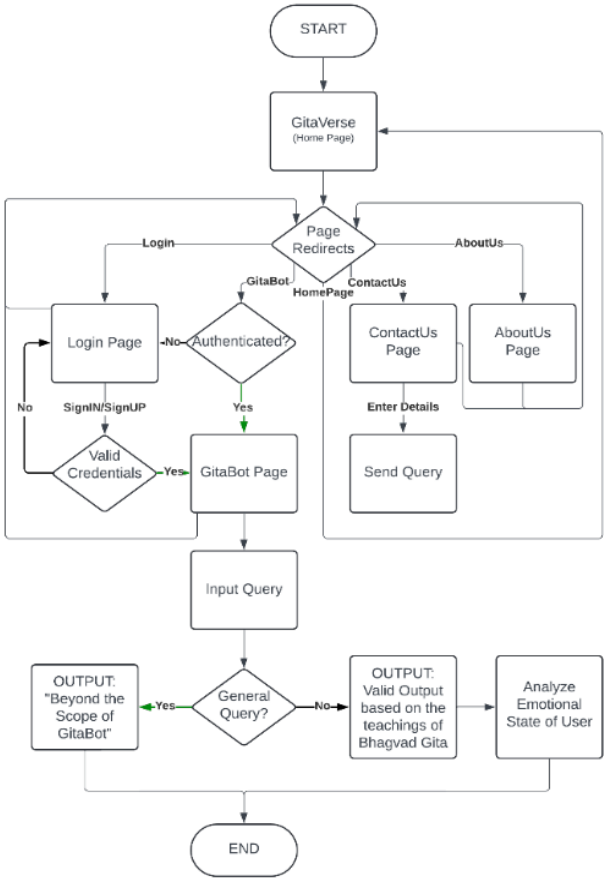
4.4 Sample Data of User Query

Table 1. User Query Sample Dataset

Emotion	Sample 1	Sample 2
Neutral	The annual report offers a thorough summary of the business's financial performance.	The recipe book contains instructions for preparing various dishes.
Surprise	Breaking: Researchers have found a new species of marine life in the ocean's depths.	I couldn't believe it when I found out I had won the lottery!
Sadness	The funeral service was a somber occasion as friends and family gathered to mourn the loss of a loved one.	After years of friendship, they drifted apart, leaving behind a sense of sadness and longing.
Joy	We're thrilled to announce the arrival of our baby boy!	The team celebrated their victory with cheers and hugs, overflowing with joy.
Fear	The eerie silence of the abandoned house sent shivers down her spine as she cautiously stepped inside.	Reports of a serial killer on the loose have left the community living in fear.
Anger	I'm outraged by the injustice of the verdict, which let the perpetrator go free.	How dare they raise prices again? It's time to boycott this company!

4.5 GitaVerse Structure

Fig. 5. Structure of GitaVerse



5. Future Work

In addition, we can integrate the chatbots as immediate help for mental health people, dealing with anxiety and depression, this can be done by taking on board the mental health professionals while having therapy and the sufferers can easily access the learning of lord Krishna by a user-friendly interface of Bhagavad Gita chatbot.

5.1 Continuous Improvements and Future Work

The aim is to make an effective and powerful chatbot that doesn't compromise with the wisdom and spirituality of the Bhagavad Gita and provides the end users the enlightenment they need to overcome the hurdles they are facing in their lives. This Bhagavad Gita chatbot is one of the most reliable tools to attain harmony and satisfaction in one's life.

1. **Data Set Expansion.** Also, to have a wiser and broader perspective, a more extensive dataset from other spiritual scriptures like 'Ram Charitmanas', 'Puranas', etc. has to be intensified so that it acquires every possible problem's solution.

2. **Formation Of Multi-Model Content.** Exploring the implementation of multi-model features, like audio and video references, taken from presently available Mahabharata TV

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shows, will eventually make the user experience over the moon, as it'll be more interactive than just reading to your solutions on chatbot. This will contribute valuable insights into the scripture of Bhagavad Gita.

3. **Collaborating With Psychiatrists.** Foster collaborations and discussions with mental health doctors and professionals will ensure more closely that the chatbot is helping the victim or not. In this way, developers will get the best feedback and can make changes according to the needs of users.

4. **Cross-Cultural Adaptation.** Making the chatbot work for users from different cultures, by ensuring that the solutions and benightment the chatbot shows culturally fit the users, as the learnings might contradict other religion's learning. This might involve considering other's cultural and religious learning too in the Gita chatbot.

5. **Engaging Users for A Larger Period.** Merging chatbots with wearable technologies, enhancing the user-friendly experience for constant monitoring of their stress levels, and delivering the soulful learnings of Bhagavad Gita whenever needed.

6. Conclusions

The development and implementation of our Bhagavad Gita chatbot has demonstrated the potential of artificial intelligence in the realm of spiritual and philosophical guidance. This chatbot, designed to answer questions through the teachings of the Bhagavad Gita, serves as a bridge between ancient wisdom and modern technology.

Our research has shown that the chatbot can effectively interpret user queries and provide insightful responses based on the teachings of the Bhagavad Gita. This not only makes the profound wisdom of the Gita more accessible but also provides a unique platform for individuals seeking guidance and answers to life's complex questions.

It is essential to understand that the chatbot is a powerful tool., it is not a replacement for personal study and reflection on the Bhagavad Gita. The chatbot serves as a supplement, aiding in understanding and interpreting the teachings of the Gita.

Further efforts could be directed towards enhancing the chatbot's capacity to comprehend and respond to more complex queries, as well as expanding its knowledge base to include other philosophical texts. (Testas, 2023) This research opens up exciting possibilities for the use of AI in the field of philosophy and spirituality, and we look forward to seeing how this technology continues to evolve.

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